

SANCHEZ

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SUMMARY

Computer Science graduate with experience spanning systems programming, web development, machine learning, and low-level programming. Actively developing Java and Spring Boot proficiency with a strong OOP foundation and cloud exposure via Microsoft Azure. Bilingual in English and Spanish.

TECHNICAL SKILLS

Languages: Java, Python, C++, C, C#, Ruby, Rust, Assembly x86

Web / Frontend: React, JavaScript, HTML, CSS, Flutter

Tools: Git, GitHub, Jira, Vim, Android Studio, Unity, Wireshark, VirtualBox

Cloud & OS: Microsoft Azure, UNIX/Linux, PowerShell, CMD

PROJECTS

Personal Portfolio Website

In Progress

React | Phaser 3 | JavaScript | Vim

- Building a portfolio site with interactive game elements using React and Phaser 3, applying front-end development skills through self-directed work outside of formal coursework.

2D Top-Down Game

In Progress

Unity | C# | AI Algorithms

- Creating a 2D top-down game in Unity using AI algorithms for enemy behavior, including custom level design and gameplay mechanics.

Efficient Polyp Detection

2025

Python | YOLOv12 | Weights & Biases

- Trained a YOLOv12 object detection model for polyp detection in endoscopic images on the Kvasir-SEG dataset, targeting early colorectal cancer diagnosis.
- Applied preprocessing, augmentation, and performance tuning to balance accuracy and inference latency; used Weights & Biases for experiment tracking.

Multiplayer Game Server & Client

2024

Python | Sockets | Multi-threading

- Built a socket-based multiplayer server for Tic-Tac-Toe, Connect 4, and Mancala with secure handshakes, multi-threaded client handling, and real-time broadcasting.
- Enhanced reliability with structured logging and efficient broadcasting for concurrent multi-user sessions.

Unix Shell Implementation

2024

C | UNIX/Linux | Process Management

- Extended a Unix shell with PATH resolution, dynamic prompts, command history, subshell nesting, and child process management via waitpid.

User-Mode Thread Scheduler

2024

C | Concurrency | OS Concepts

- Implemented cooperative FIFO and Round-Robin schedulers in user space with semaphore-based synchronization and mutual exclusion.

Ping Pong

2024

Assembly x86 | Low-Level Programming

- Developed a two-player ping pong game in x86 Assembly, implementing game logic, collision detection, and performance optimization at the hardware level.

Digital Systems — Ripple-Carry Adder

2023

Logic Gates | Multiplexers | Circuit Design

- Assembled a digital circuit using logic gates and multiplexers to add two 3-bit binary numbers, implementing a ripple-carry adder with overflow detection.

Grades Management System

2022

Ruby on Rails | Full-Stack | Authentication

- Built a full-stack web app with CRUD operations, secure user authentication, and role-based access control for student grade management.

EDUCATION

University of Texas Rio Grande Valley

May 2025

Bachelor of Computer Science | GPA: 2.91 | Minor: Asian Studies (GPA: 3.50)

South Texas College

May 2016

Associate of Arts, Interdisciplinary Studies | GPA: 3.32

AWARDS & CERTIFICATIONS

- IBM Z 101 Certificate
- UTRGV Engaged Scholar Symposium — Honorary Mention & Distinguished Scholar Award (2018) for performance: "Tae Kwon Do - Korean Martial Arts"